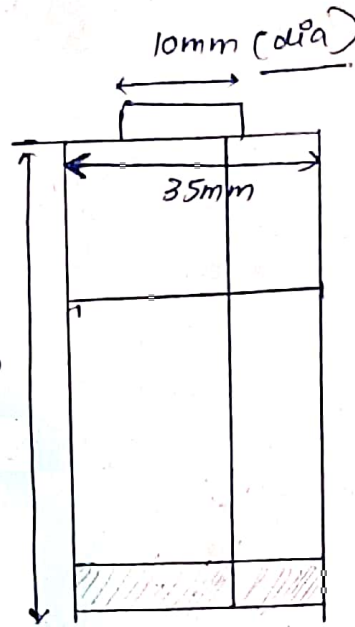
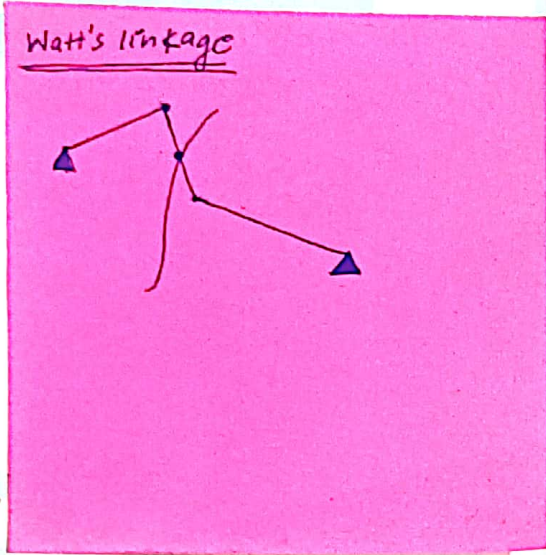


190mm x 100mm

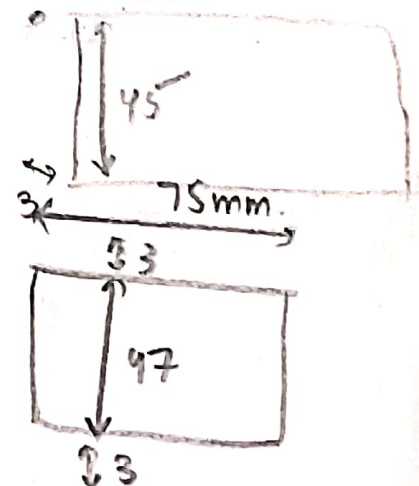
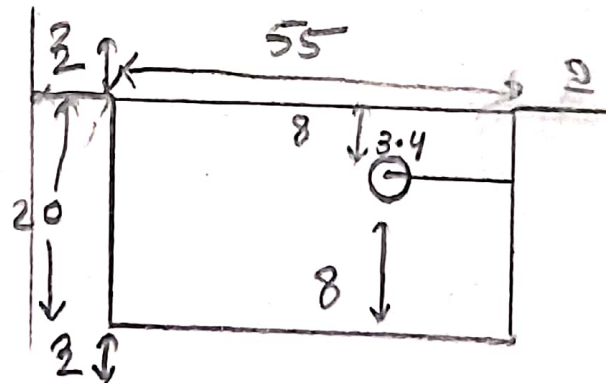
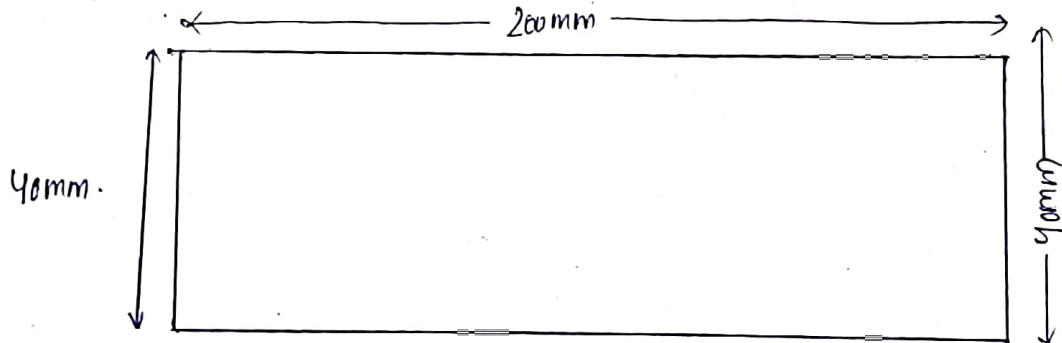
~~200mm x~~

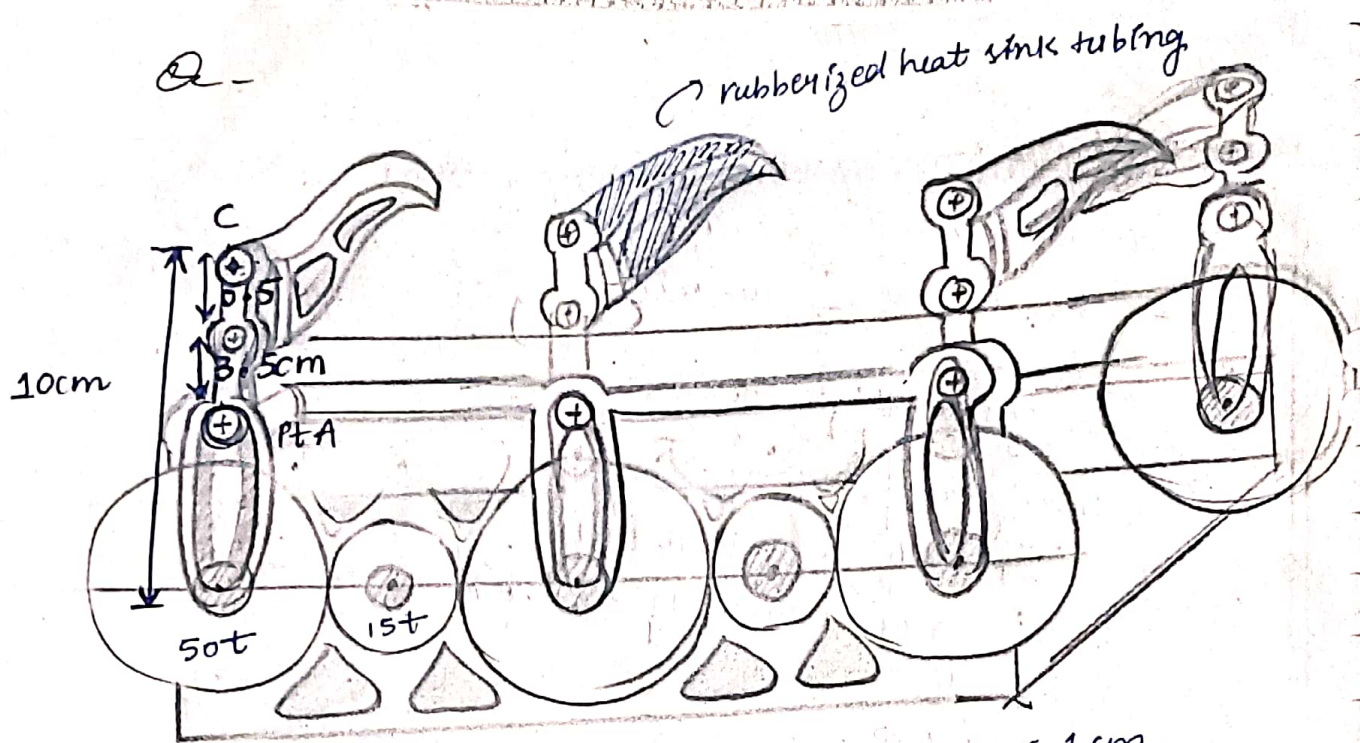
65mm



35mm

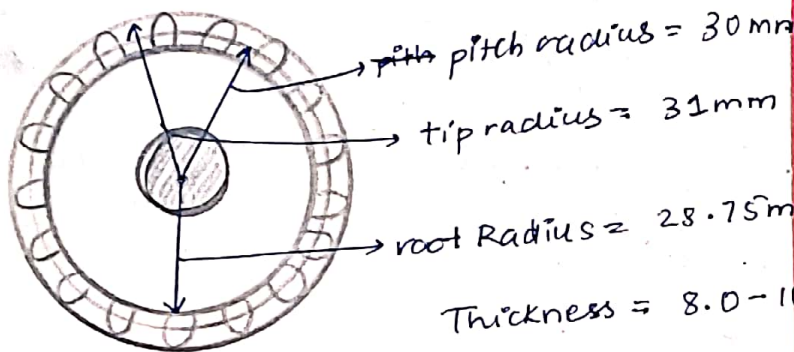
95mm dia



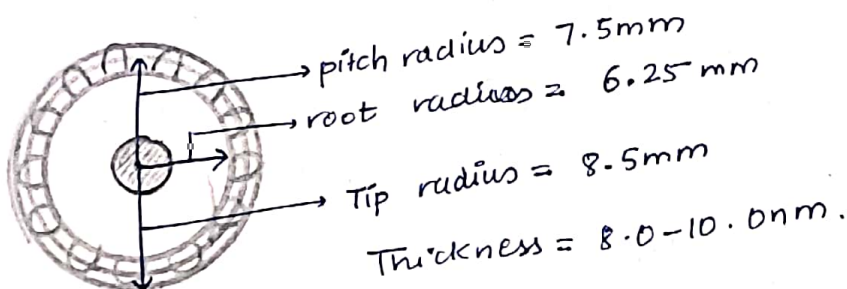


max weight = 500gm
max length = 25cm

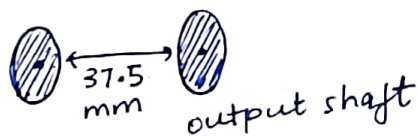
diameter of rope < 1cm.



larger gear



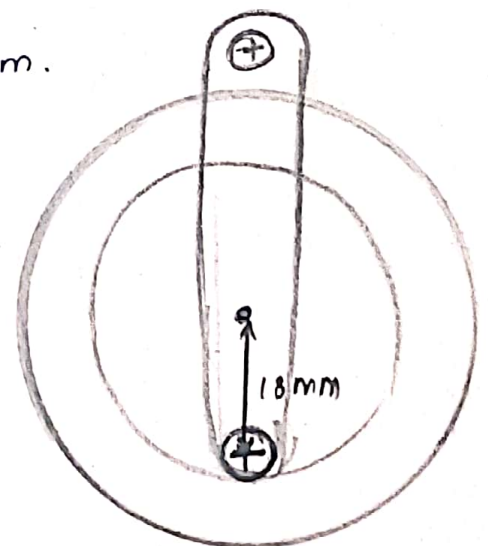
shaft spacing :- motor shaft



15t (small) and 60t (large)
Reduction ratio: $\frac{60}{15} \times \frac{60}{15} = 16:1$
output RPM: $100 \div 16 = 6.25$ RPM
gear ratio: $\frac{60}{15} = 4:1$
90SEC $\rightarrow$ 5.4cm stride
crank radius: pivot point 18mm from the center of each large gear

$$7.5 \times 3 = 22.5 \text{ mm}$$

2.25 cm



driver gear
8 teeth

driven gear
40 teeth

$$\text{gear ratio} = \frac{40}{8} = 5$$

driver gear
24 teeth

driven gear
40 teeth

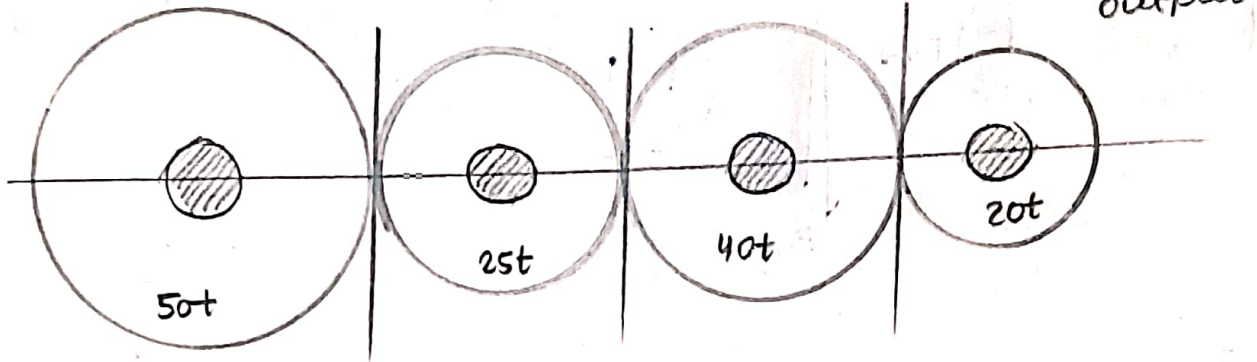
$$\text{gear ratio} = \frac{40}{24} = 1.67$$

input torque $\Rightarrow$ —

output torque $\Rightarrow$ initial torque $\times$ gear ratio
--

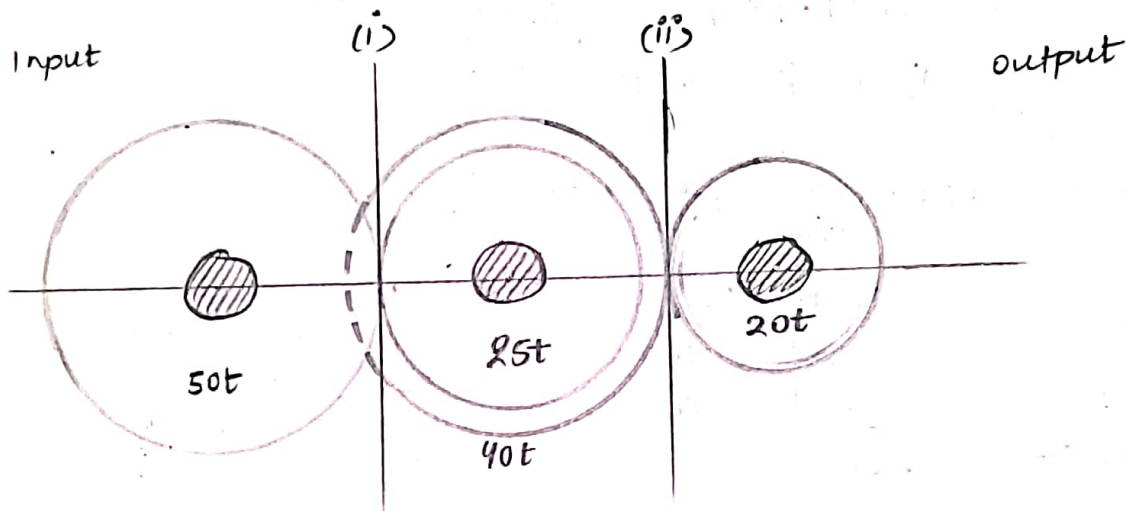
compound gears : Gear ratio

Input



$$\text{Gear ratio} = \frac{20t}{50t} = 0.4$$

Input

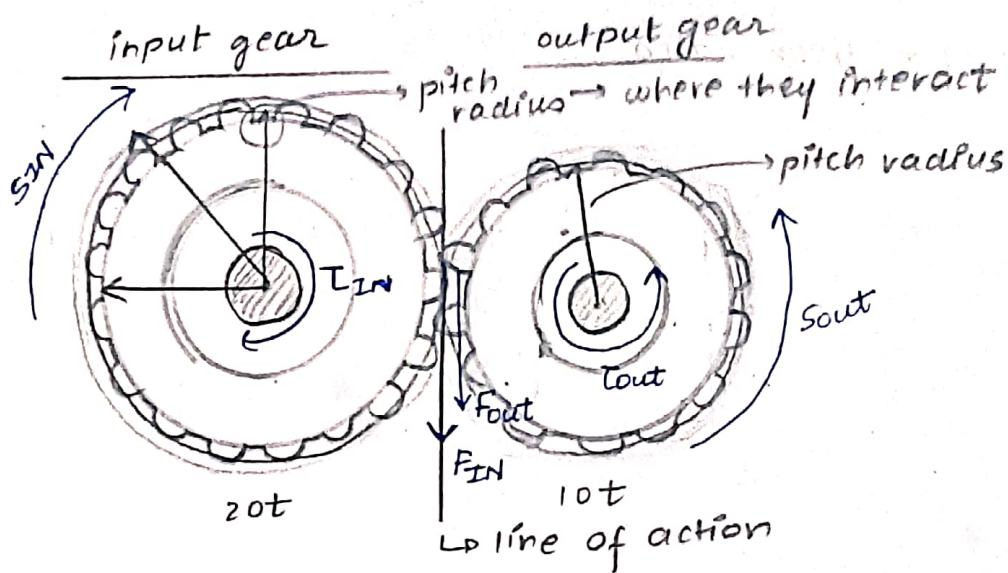


$$\text{gear ratio (i)} = \frac{25t}{50t} = 0.5$$

$$\text{gear ratio (ii)} = \frac{20t}{40t} = 0.5$$

$$\text{gear ratio (total)} = 0.5 \times 0.5 = \underline{0.25}$$

simple gear (spur gear).



Torque :- $T = Fr \sin \theta$
 $T_{in} = F_{in} r_{in}$ (pitch radius of input gear)
 $T_{out} = F_{out} r_{out}$ (pitch radius of output gear)

$$F_{in} = \frac{T_{in}}{r_{in}}, \quad F_{out} = \frac{T_{out}}{r_{out}}$$

$$F_{out} = F_{in}$$

$$\frac{T_{out}}{r_{out}} = \frac{T_{in}}{r_{in}}$$

Rotation :- $S = \theta r$

$$S_{in} = \theta_{in} r_{in}$$

$$S_{out} = \theta_{out} r_{out}$$

$$S_{in} = S_{out}$$

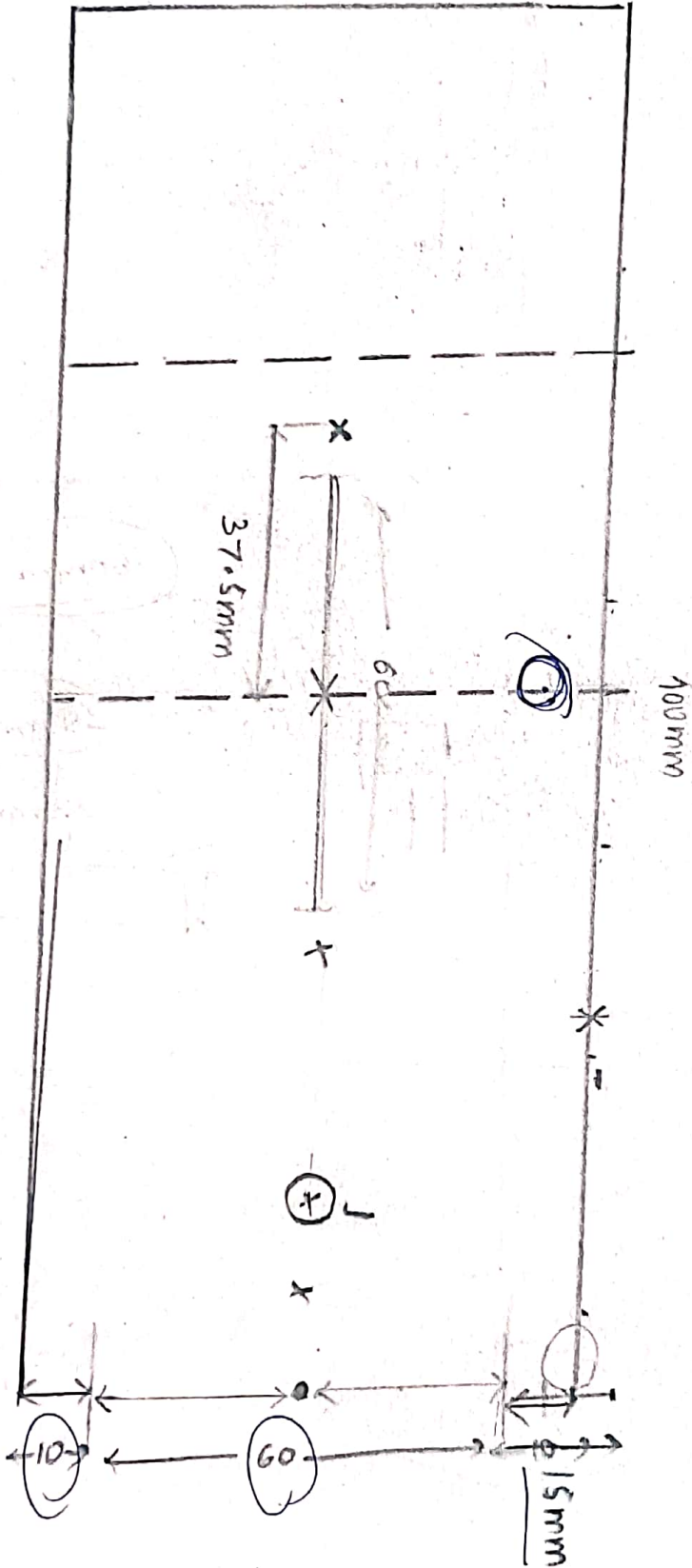
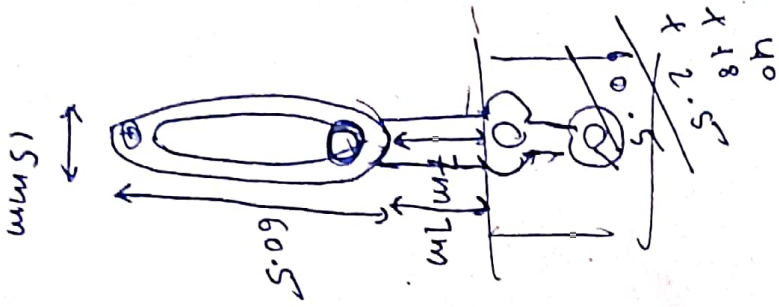
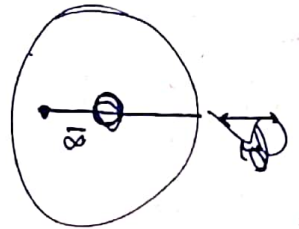
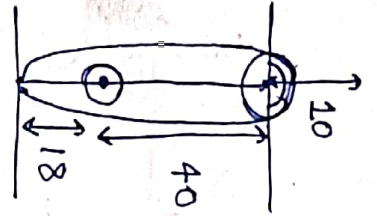
$$\theta_{in} r_{in} = \theta_{out} r_{out}$$

$$\frac{r_{out}}{r_{in}} = \frac{\theta_{in}}{\theta_{out}}$$

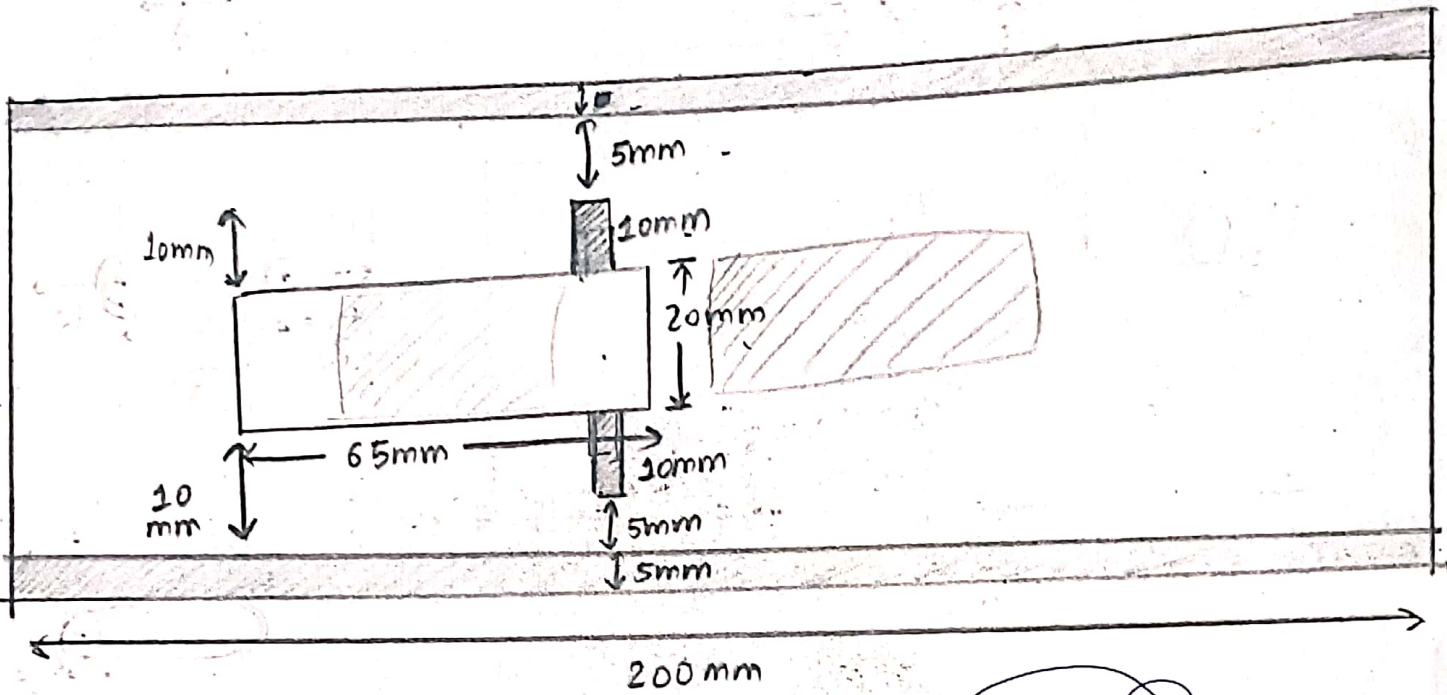
gear ratio :-

$$GR = \frac{r_{out}}{r_{in}} = \frac{n_{out}}{n_{in}} = \frac{T_{out}}{T_{in}} = \frac{\theta_{in}}{\theta_{out}} = \frac{\omega_{in}}{\omega_{out}}$$

$$\text{work} = T\theta = \tau\theta$$



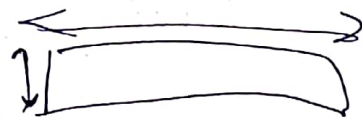
$$\frac{20}{15} = 85$$



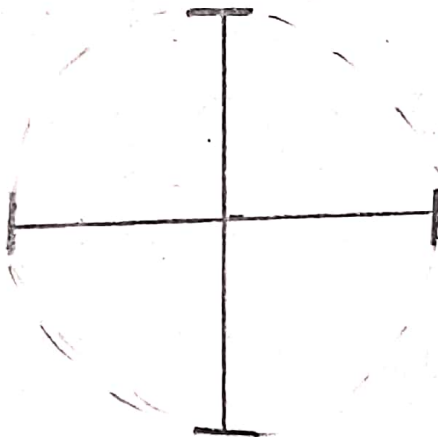
60mm

25mm

2.5mm



10
4 x coupling
10
10



Gear (4) and (5)

backlash = 0.15mm

module = 1.0

Pressure Angle = 20°

Number of teeth = 60 mm

Root fillet radius = 0.40mm

Gear thickness = 10mm

Hole diameter = 3.40 mm

pitch diameter = 60.0mm.

{ Date - 2-3-16
Invoice - 1216
DLILAR0836 }

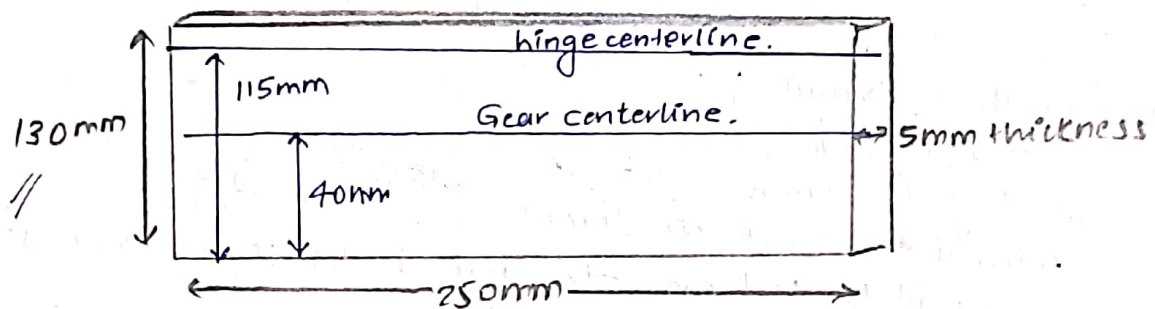
Sup. ref.

Buyer order no. -

Main Body (side plate dimensions)

length = 250mm
height = 130mm
Thickness = 5mm.

2.5



Base body

